

Intermediate Microeconomics

Week 2 · Day 1 — Supply and Demand (Perloff, Ch. 2)

Main text: Perloff, *Intermediate Microeconomics*, Ch. 2

Where we are — Week 2 plan

Week 1 built the *tools*: functions, constrained optimisation, marginal reasoning. Week 2 puts them to work on markets.

The week at a glance

- Day 1** Supply, demand, equilibrium, comparative statics, elasticities
- Day 2** Welfare: consumer/producer surplus, price controls, tax incidence, DWL
- Day 3** The competitive firm and market: entry, exit, long-run supply
- Day 4** Market power: monopoly and monopsony

A single market ($Q_d = 120 - 2P$, $Q_s = -60 + 4P$) will follow us all week: today we solve it, Day 2 we tax it, Day 3 we zoom into one of its firms, Day 4 we ask what happens if one firm owns it all.

Learning goals — Day 1

By the end of today you should be able to:

- ➊ Distinguish a *movement along* a curve from a *shift of* the curve, and state the ceteris-paribus content of each.
- ➋ Solve for market equilibrium algebraically and read it off a diagram.
- ➌ Sign and *size* the equilibrium response to demand and supply shocks (comparative statics).
- ➍ Compute point elasticities of demand and supply, and explain why elasticity varies along a linear demand curve.
- ➎ Link elasticity to revenue and to the *incidence* of a shock — the bridge to Day 2.
- ➏ State the assumptions under which the supply-and-demand model is the right tool.

The demand curve

A *demand function* gives quantity demanded as a function of price and shifters:

$$Q_d = D(P, Y, P_s, P_c, \tau, \dots)$$

where Y is income, P_s/P_c prices of substitutes/complements, τ tastes.

Movement vs. shift

- Change in **own price** $P \Rightarrow$ *movement along* the curve.
- Change in **any shifter** $\Rightarrow$ *shift of* the whole curve.

Our working demand curve (holding shifters fixed):

$$Q_d = 120 - 2P \quad \Longleftrightarrow \quad P = 60 - \frac{1}{2}Q \quad (\text{inverse demand}).$$

Slope $dQ_d/dP = -2 < 0$: the law of demand.

The supply curve

A *supply function* gives quantity supplied as a function of price and cost shifters:

$$Q_s = S(P, w, r, \text{technology}, \dots).$$

Our working supply curve:

$$Q_s = -60 + 4P \quad \Longleftrightarrow \quad P = 15 + \frac{1}{4}Q \quad (\text{inverse supply}).$$

- Slope $dQ_s/dP = 4 > 0$.
- The intercept $P = 15$ is the *choke price*: below \$15 nothing is supplied. Keep this number — on Day 3 it reappears as the firm's shut-down price (min AVC).

Interpretation preview

Supply is (the horizontal sum of) firms' marginal-cost curves. $P = 15 + \frac{1}{4}Q$ is $n = 20$ identical firms each with $MC = 15 + 5q$. We derive this on Day 3.

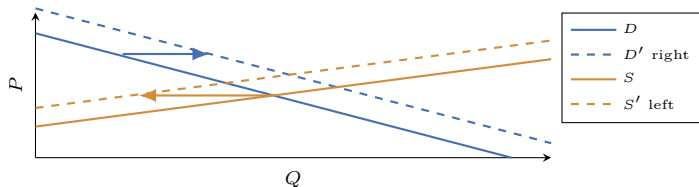
What shifts the curves?

Demand shifters (move D)

Factor	Shift
Income $Y \uparrow$	right (normal) / left (inferior)
Substitute price $\uparrow$	right
Complement price $\uparrow$	left
Taste for good $\uparrow$	right
# buyers $\uparrow$	right
Expected future $P \uparrow$	right (buy now)

Supply shifters (move S)

Factor	Shift
Input price $w, r \uparrow$	left
Better technology	right
# sellers $\uparrow$	right
Per-unit tax $\uparrow$ / subsidy $\uparrow$	left / right
Expected future $P \uparrow$	left (hold back)
Bad weather (agriculture)	left



Own price is on *neither* list — a change in P is a *movement along*, never a shift. Every factor above relocates a whole curve and, through the crossing, moves (P^*, Q^*) .

Equilibrium: definition and solution

Equilibrium price P^* clears the market:

$Q_d(P^*) = Q_s(P^*)$. **Step 1 — set quantities**

equal

$$120 - 2P = -60 + 4P$$

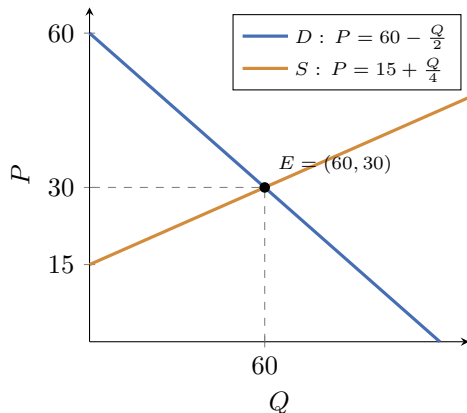
Step 2 — solve for P^*

$$180 = 6P \Rightarrow \boxed{P^* = 30}$$

Step 3 — back-substitute for Q^*

$$Q^* = 120 - 2(30) = \boxed{60}.$$

Numeric check: $Q_s = -60 + 4(30) = 60 = Q_d$. ✓



Why the market converges to E

- At $P = 36 > P^*$: $Q_d = 120 - 72 = 48$, $Q_s = -60 + 144 = 84$. **Excess supply** of 36 $\Rightarrow$ price is bid down.
- At $P = 24 < P^*$: $Q_d = 72$, $Q_s = 36$. **Excess demand** of 36 $\Rightarrow$ price is bid up.

Only at $P^* = 30$ is there no pressure. (Hold on to those ± 36 gaps — Day 2 turns exactly these prices into a price floor and a price ceiling.)

Existence & uniqueness (linear case)

With a strictly downward-sloping D and strictly upward-sloping S that cross in the positive quadrant, equilibrium exists and is unique. Non-existence needs pathologies (e.g. S everywhere above D).

Comparative statics — a demand shock

A rise in income shifts demand out (a normal good). New demand: $Q'_d = 150 - 2P$.

Step 1 — new equilibrium condition

$$150 - 2P = -60 + 4P$$

Step 2 — solve

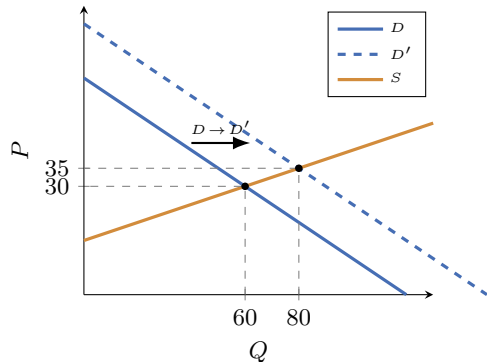
$$210 = 6P \Rightarrow P' = 35, \quad Q' = 150 - 70 = 80.$$

Step 3 — report the changes

$$\Delta P = +5, \quad \Delta Q = +20.$$

Both rise: the signature of a demand-driven shock.

Numeric check: $Q_s = -60 + 4(35) = 80 = Q'_d$. ✓



The general comparative-statics rule

Write demand and supply as $Q_d = a - bP$,
 $Q_s = -c + dP$. Equilibrium:

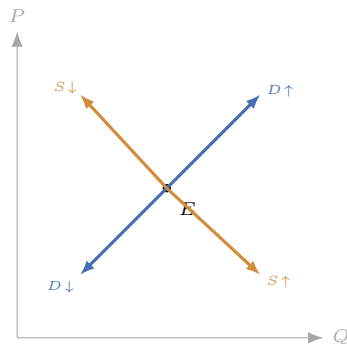
$$P^* = \frac{a + c}{b + d}, \quad Q^* = \frac{ad - bc}{b + d}.$$

Signs at a glance

Single shock	ΔP^*	ΔQ^*
Demand right ($a \uparrow$)	+	+
Demand left ($a \downarrow$)	-	-
Supply right ($c \uparrow$)	-	+
Supply left ($c \downarrow$)	+	-

Sizes depend on the slopes b, d : a steep (inelastic) supply throws more of a demand shock into price than quantity — the “how much” question elasticities answer next.

Where E moves



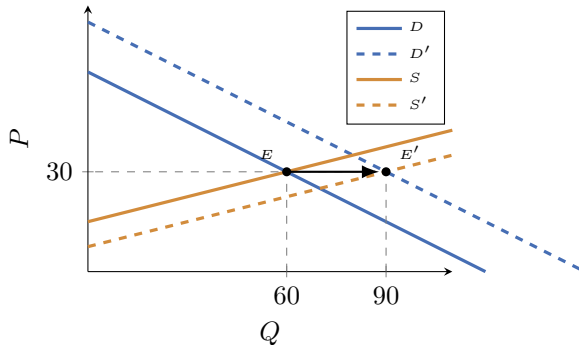
Demand shifts move P, Q together; supply shifts move them opposite ways.

When both curves move: one effect is ambiguous

If D and S shift together, the direction they agree on is determinate; the other depends on the *relative sizes* of the shifts.

The four combinations

	ΔQ^*	ΔP^*
D right, S right	+	?
D left, S left	-	?
D right, S left	?	+
D left, S right	?	-



Both shift right below, so Q^* rises for sure. Here the price pressures cancel:
 $150 - 2P = -30 + 4P \Rightarrow P^* = 30$ (unchanged),
while $Q^* : 60 \rightarrow 90$.

Shift either curve a little more and P^* moves — its sign is an *empirical* question about magnitudes, not something theory settles.

Point elasticity of demand

The price elasticity of demand is the *unit-free* responsiveness

$$\varepsilon = \frac{\% \Delta Q_d}{\% \Delta P} = \frac{dQ_d}{dP} \cdot \frac{P}{Q_d}.$$

For $Q_d = 120 - 2P$ we have $dQ_d/dP = -2$, so at the equilibrium $(P, Q) = (30, 60)$:

$$\varepsilon = (-2) \cdot \frac{30}{60} = \boxed{-1} \quad (\text{unit elastic}).$$

Supply elasticity at the same point:

$$\eta = \frac{dQ_s}{dP} \cdot \frac{P}{Q_s} = (4) \cdot \frac{30}{60} = \boxed{+2}.$$

Numeric check: $|\varepsilon| < \eta$: supply is more responsive here — flag it for tax incidence on Day 2.

Elasticity varies along a linear demand curve

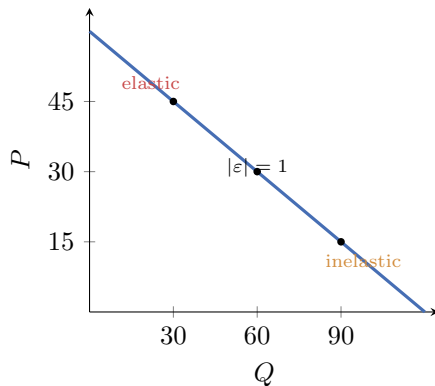
Same slope -2 everywhere, but P/Q changes:

$$\varepsilon(P) = (-2) \frac{P}{120 - 2P}.$$

P	Q	ε	region
45	30	-3	elastic
30	60	-1	unit elastic
15	90	$-\frac{1}{3}$	inelastic

Rule for *any* linear

demand: the midpoint of the price-intercept is unit elastic; above it elastic, below it inelastic.



Elasticity and revenue

Revenue $R = P \cdot Q(P)$. Differentiate:

$$\frac{dR}{dP} = Q + P \frac{dQ}{dP} = Q(1 + \varepsilon).$$

Reading

- $|\varepsilon| < 1$ (inelastic): $dR/dP > 0$ — raise price to raise revenue.
- $|\varepsilon| > 1$ (elastic): $dR/dP < 0$ — cut price to raise revenue.
- $|\varepsilon| = 1$: revenue is stationary — *maximised*.

Here revenue peaks at $P = 30$, $R = 30 \times 60 = 1800$, precisely where $\varepsilon = -1$. This $Q(1 + \varepsilon)$ object returns on Day 4 as marginal revenue — monopoly lives on the elastic side.

Two more elasticities, and a constant-elasticity form

Income and cross-price

$$\varepsilon_Y = \frac{dQ}{dY} \frac{Y}{Q} \quad (\text{normal } >0, \text{ inferior } <0), \quad \varepsilon_{xy} = \frac{dQ_x}{dP_y} \frac{P_y}{Q_x} \quad (\text{subs } >0, \text{ comp } <0).$$

Constant-elasticity demand

A power (log-linear) form has the *same* elasticity at every price:

$$Q = AP^\alpha \Rightarrow \varepsilon = \frac{dQ}{dP} \frac{P}{Q} = \alpha AP^{\alpha-1} \cdot \frac{P}{AP^\alpha} = \alpha.$$

Taking logs, $\ln Q = \ln A + \alpha \ln P$ — α is the slope of a log-log regression. This is why empirical demand is usually estimated in logs.

When is supply-and-demand the right tool?

The model assumes a *perfectly competitive* market:

- ① many buyers and many sellers — each a price taker;
- ② a homogeneous product;
- ③ free entry and exit (in the long run);
- ④ good information; low transaction costs.

When it misleads

Concentrated sellers (monopoly, Day 4), differentiated products, search/matching frictions, or important externalities (Day 2 hints). Then a single D/S crossing is no longer the whole story.

Challenge solution — a cost-saving innovation

Question. A process innovation lowers unit cost, shifting supply out to $Q'_s = -30 + 4P$ (choke price falls $15 \rightarrow 7.5$). Predict the new price and quantity, and say who gains.

Step 1 — new equilibrium

$$120 - 2P = -30 + 4P \Rightarrow 150 = 6P \Rightarrow P' = 25.$$

Step 2 — quantity

$$Q' = 120 - 2(25) = 70.$$

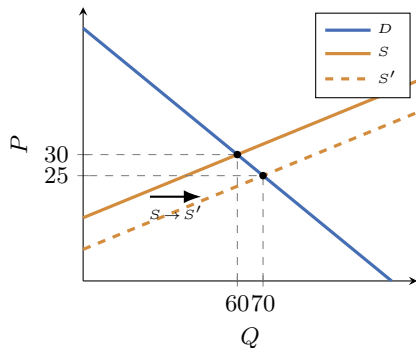
Step 3 — changes and incidence

$$\Delta P = -5, \quad \Delta Q = +10.$$

Price falls, quantity rises — the supply-shock signature (opposite of the demand shock earlier).

Numeric check: $Q'_s = -30 + 4(25) = 70 = Q_d$. ✓

Who gains? Consumers pay less and buy more; whether producers gain depends on the elasticities — we make this precise with surplus on Day 2.



Day 1 checkpoint

You can now

- solve $Q_d = Q_s$ for (P^*, Q^*) and read it off a diagram;
- sign *and* size a shock: demand out $\Rightarrow (P \uparrow, Q \uparrow)$; supply out $\Rightarrow (P \downarrow, Q \uparrow)$;
- compute point elasticities and locate the unit-elastic point;
- connect elasticity to revenue via $dR/dP = Q(1 + \varepsilon)$.

Carried into Day 2

the market $(30, 60)$, the choke prices 15 and 7.5, and the fact that $\eta > |\varepsilon|$ here — which will decide who bears tomorrow's tax.

In-class exercises — Day 1

Time for the in-class exercises.

- **Exercise 1** — a fresh market: equilibrium and a supply shock (comparative statics).
- **Exercise 2** — elasticity: point elasticity along a linear curve, the revenue-maximising price, and a constant-elasticity form.

Work in pairs; we reconvene to compare the sign *and size* of each answer.